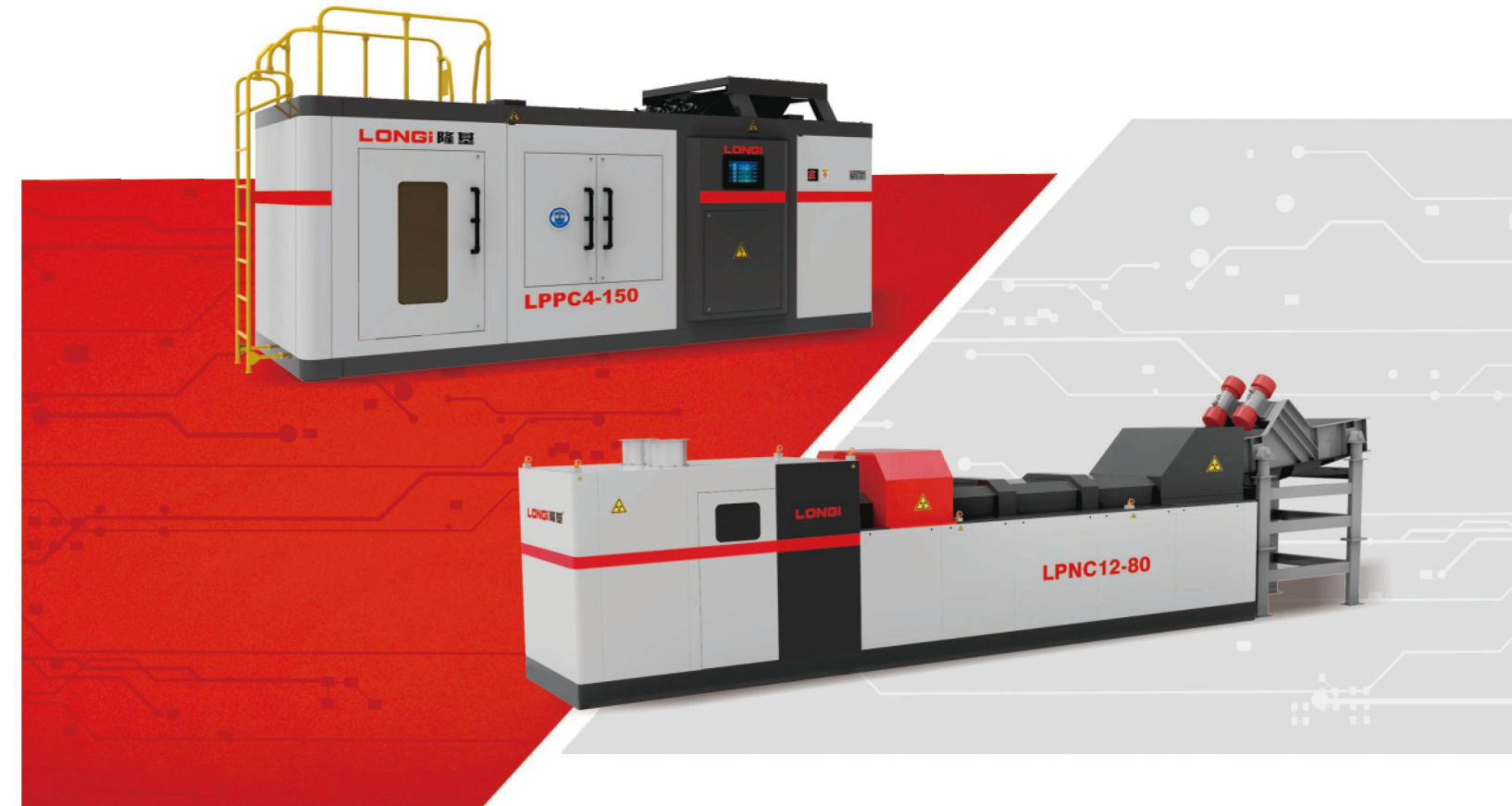




**MAXIMIZING VALUE WITH HIGHLY
EFFICIENT PRE-CONCENTRATION**



LPPC & LPNC

**LONGi Intelligent Optronics
Sorter**

R&D ABILITY



8000

Total investment value
RMB 80 million

30%

Proportion of technical
R & D personnel

LONGi R&D center is found in 2011, it is the biggest facility in the magnetic equipment industry with an 80 million RMB investment and 10,000 square meters area.

LONGi's R&D center has the ability of testing dry separation, wet separation, gravity separation and flotation. This improves the existing products' design and processes flow while developing and designing new products.

LONGi has more than 300 units of forty different types of experimental and testing equipment and devices. These are used to test and prove the operability and reality of our new technology and are developed to ensure the reality of the tests and the quality of products. R&D is a reliable foundation for LONGi's product and service.

The development mission of LONGi's R&D Center is to adhere to the customer demand. By researching and developing new products and new technologies. LONGi integrates and cooperates with major scientific research facilities along with a group of top national R&D teams. In doing so, LONGi has obtained numerous influential and major technological achievements.

National Laboratory



Chemical Analysis



Dry Separation Test



Magnetic Separation Test



Atomic Absorption Analysis



Grinding Test



Sample Preparation



Granularity Analysis



Sample Storage



Super Iron Ore Concentrate Processing Pilot Line



Grinding Test



Multi Levels Shelf Storage



Cooperation With The Northeastern University

TECHNICAL SPECIFICATIONS

Sensor

Using the difference in physical properties of substances, the coresponding sensors are used to measure their physical properties, and then the measured data is analyzed and processed with the help of computers. Finally, various actuators are used to achieve theseparation of useful substances and waste.

X-Ray Sorting

When minerals interact with electromagnetic waves of different wavelengths, they will have different characteristics such as absorption, scattering or reflection. At present, the separation methods of ore include radioactive separation method, fluorescent separation method, X-ray absorption method, ultraviolet fluorescence method, photoelectric method, infrared method and so on. Among various sorting methods, photoelectric sorting and X-ray sorting are mainly used.

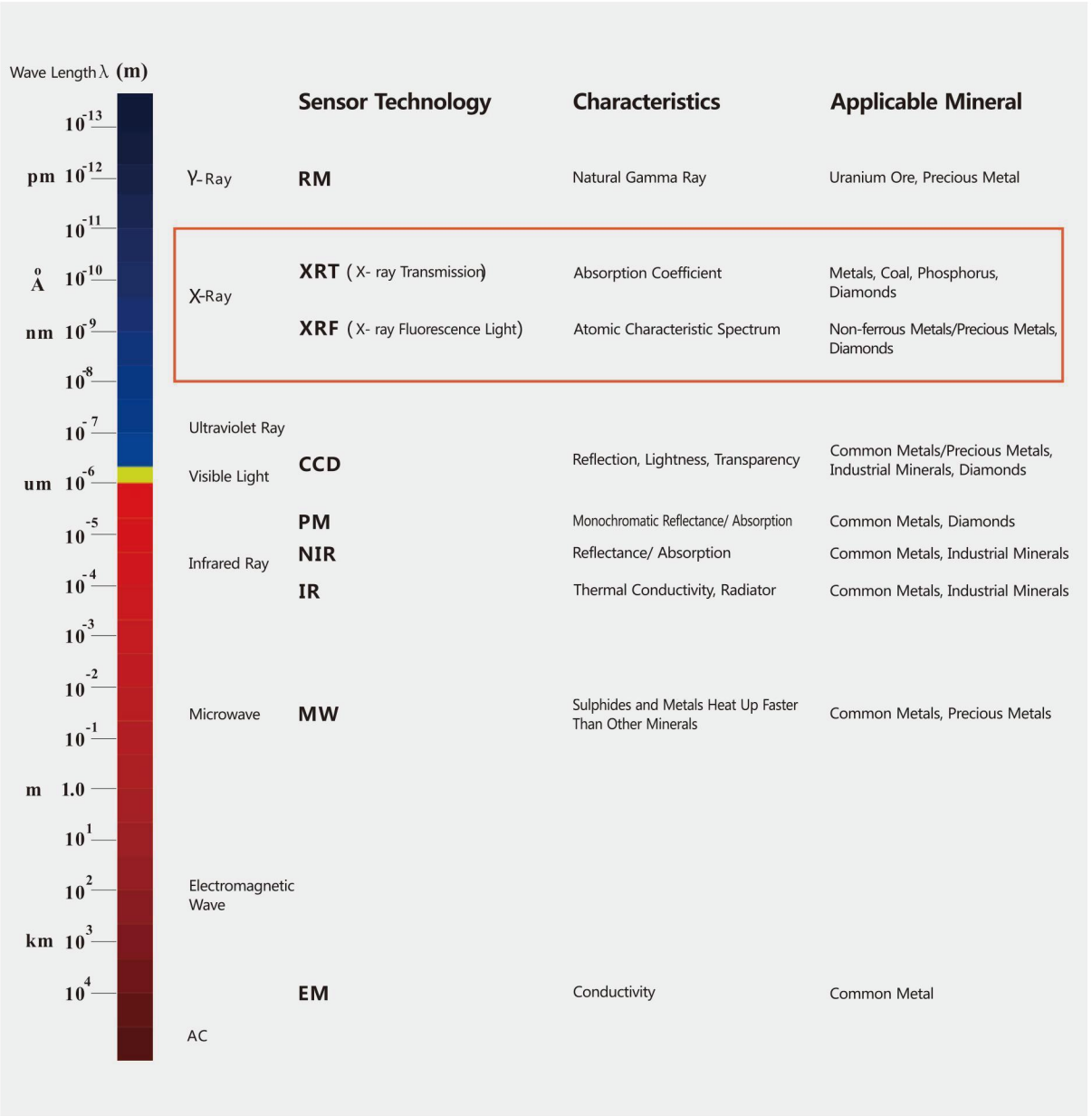
According to X-rays irradiating ores with different characteristic reactions, X-ray sorting methods are divided into X-ray fluorescence method, X-ray reflection method, X-ray absorption method and so on.



Target element range: 20Ca~92U

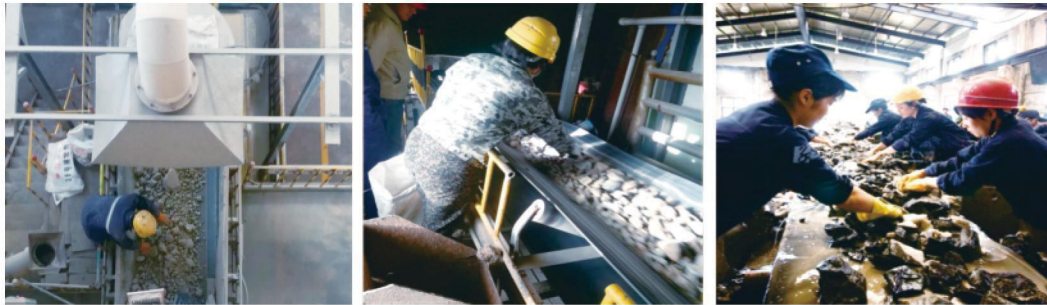
Material Characteristics And Applicable Material

At present, X-rays are used to separate the gangue from the valuable non-ferrous metal ores by the different absorbing effects or the fluorescence effects produced by X-Ray irradiation which then are processed through a computer to analyze the data calculation and separate them by mechanical devices.



POSITIONING

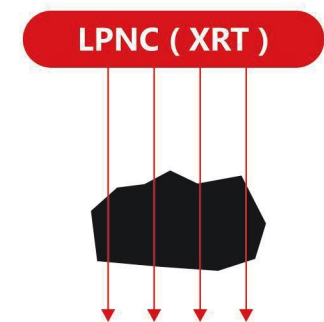
Completely Replaces Manual Sorting



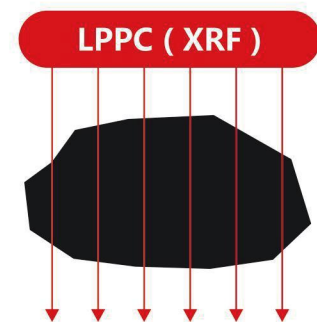
The mining depletion rate has increased as mining has been developed over the years resulting in increased energy consumption and material consumption in subsequent crushing, grinding, and sorting processes, resulting in increased production costs. Therefore, pre-sorting is an effective way to improve the production efficiency and reduce the production cost.

Discards Waste Before Grinding

Based on the current industry situation, in order to reduce costs and increase economic benefits as much as possible, LONGi began to conduct research and investigations on domestic non-ferrous metal mines in 2012, and increased its research and development efforts on waste disposal projects before grinding and to create the target of, "Realized 100% waste disposal before grinding by multi-steps sorting" .



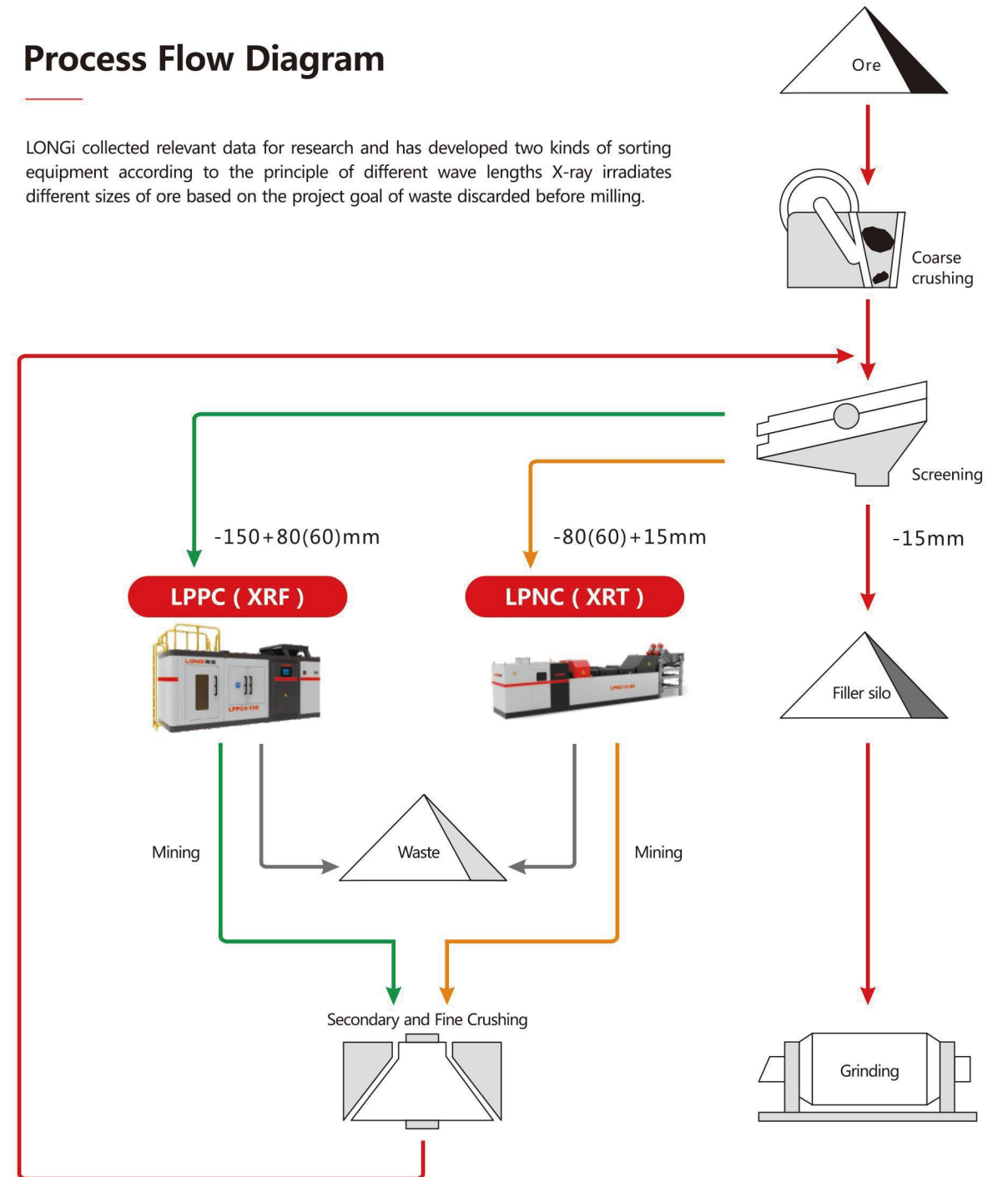
Waste discarding for material size range of -80+15mm



Waste discarding for material size range of -150+60mm

Process Flow Diagram

LONGi collected relevant data for research and has developed two kinds of sorting equipment according to the principle of different wave lengths X-ray irradiates different sizes of ore based on the project goal of waste discarded before milling.



LPNC



INTELLIGENT SORTING MACHINE LPNC(XRT)

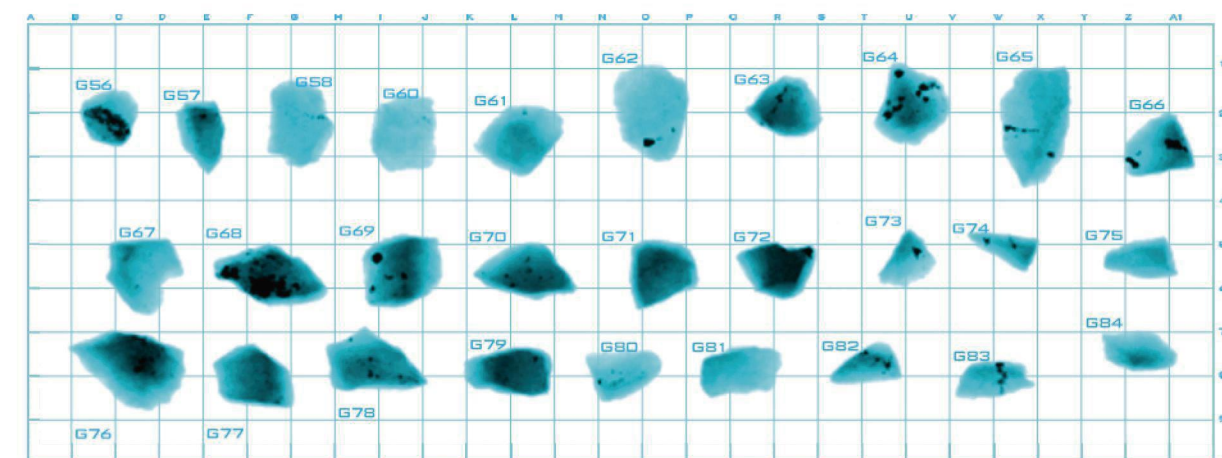
Application

- Application scope:** It is mainly applied for pre-selection of non-ferrous ore which contains gold, silver, copper, zine, wolfram etc.
- Application position:** It is applied for the stage of after secondary and fine crushing before grinding, to increase the grade before grinding and decrease the feeding quantity for grinding.
- Application range:** Greenfield project, renovation of old plant, reuse of the low-grade ore.
- Application size:** -80+15mm
- Advantage:** High-precision AI intelligent algorithm, X-ray signal deep recognition network, the identification on all range of the ores, and accurate detection of the components of ores.

Features

- Larger capacity:** larger capacity than that of traditional waste discarding processes.
- Lower consumption:** Lower power consumption, only about 0.2~0.3kw/h per ton.
- Easy operation and management:** it does not need special workers to operate the equipment after the equipment is set up. The monitoring system can have a centralized control and automatically facilitate management.
- Higher modularity level:** Modular production on all of the components of the equipment. Easy for maintenance, spare parts inventory and management.
- High reliability:** The equipment is equipped with many kinds of international well-known branded parts, which guarantees a lower failure rate.
- High waste discarding efficiency:** Compared with the traditional discarding process, the LPNC can sort smaller well liberated granularity by higher discarding efficiency.

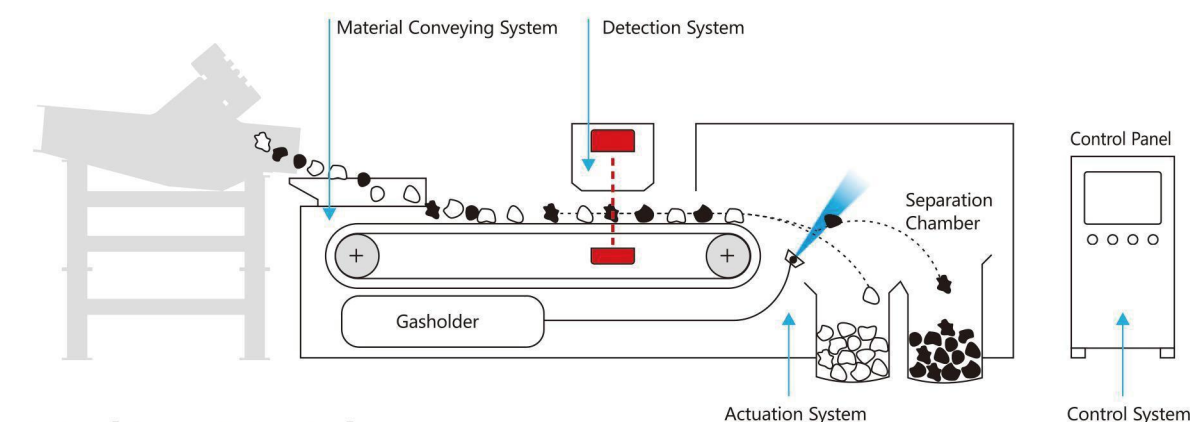
X-ray Image



Equipment Structure

The LPNC sorter is composed by four parts: a material conveying system, a detection system, a control system and an actuation sorting system.

The material conveying system is composed of the vibration feeder and the conveyor belt; the detection system is composed of the X-ray source and the detectors; the electronic control part is the control system of the equipment; and the actuation system is composed by the pneumatic driven parts.



Working Principle

The material is fed onto the conveyor belt evenly by use of a vibratory feeder. The material passes through the detection area at a uniform speed on the conveyor belt while the detection system transmits the material information to the control system.

The control system's computer receives and analyze the signals allowing it to instruct the actuation system to separate the material.

The positive selection or rejection are based on the actual process request which helps to improve the working efficiency.



INTELLIGENT SORTING MACHINE LPPC(XRF)

Application

Application scope: It is mainly applied for pre-selection of precious metals, non-ferrous ores and other rare ore that contain gold, silver, molybdenum, copper, zine, nickel, wolfram, lead and zinc , vanadium etc.

Application position: It is normally applied to the stage after crushing and before the ball mill, in order to improve the grade of the ore to the mill.

Application range: Green field plant, renovation of old plant, reuse of low-grade ore.

Application size: -150+60mm

Advantage: It can replace manual sorting completely. It is a dry pre-selection process which can discard 20%~50% of the waste in advance, improving grade before processing. This saves energy and is environmentally friendly.



Gold Ore



Silver Ore



Molybdenum Ore



Copper Ore



Nickel Ore



Wolfram Ore



Lead And Zinc Ore



Vanadium Ore



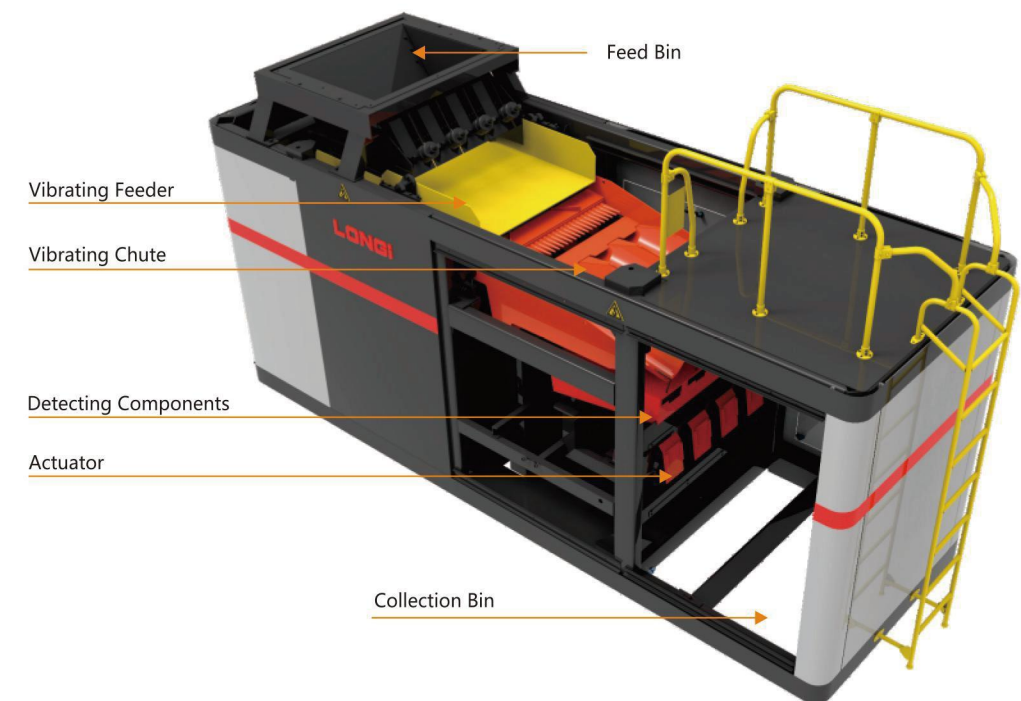
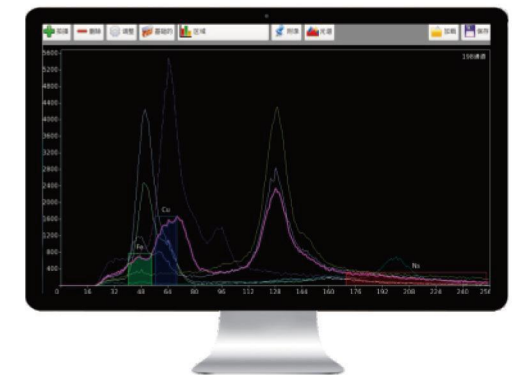
Antimony Ore



Fluorite Ore

Technology Features

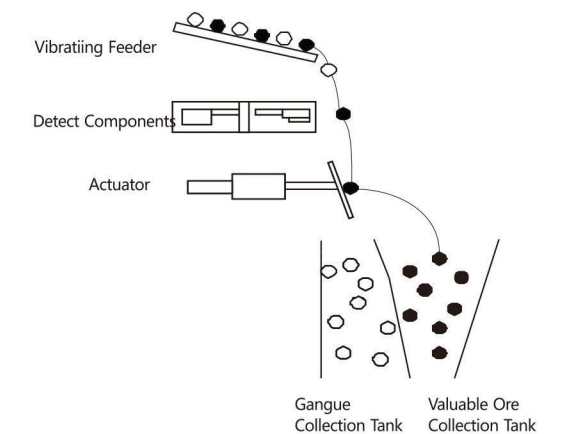
- 1、Automatic pre-selection completely replaces manual sorting at a highly efficient level.
- 2、The controls are centralized for easy operation and less manpower.
- 3、It discards the waste in advance which helps to improve the feed grade being processed and reduces the amount of crushing required.
- 4、Pre-sorting helps to reduce the amount of crushing and simplifies the downstream processing as well as reduces the amount that goes to the tailing dam which extends the life time of the tailings dam.



Working principle

The main machine is composed by a feeding bin, a vibratory feeder, a vibratory chute, the detection assembly, an actuator, the collection bin, etc.

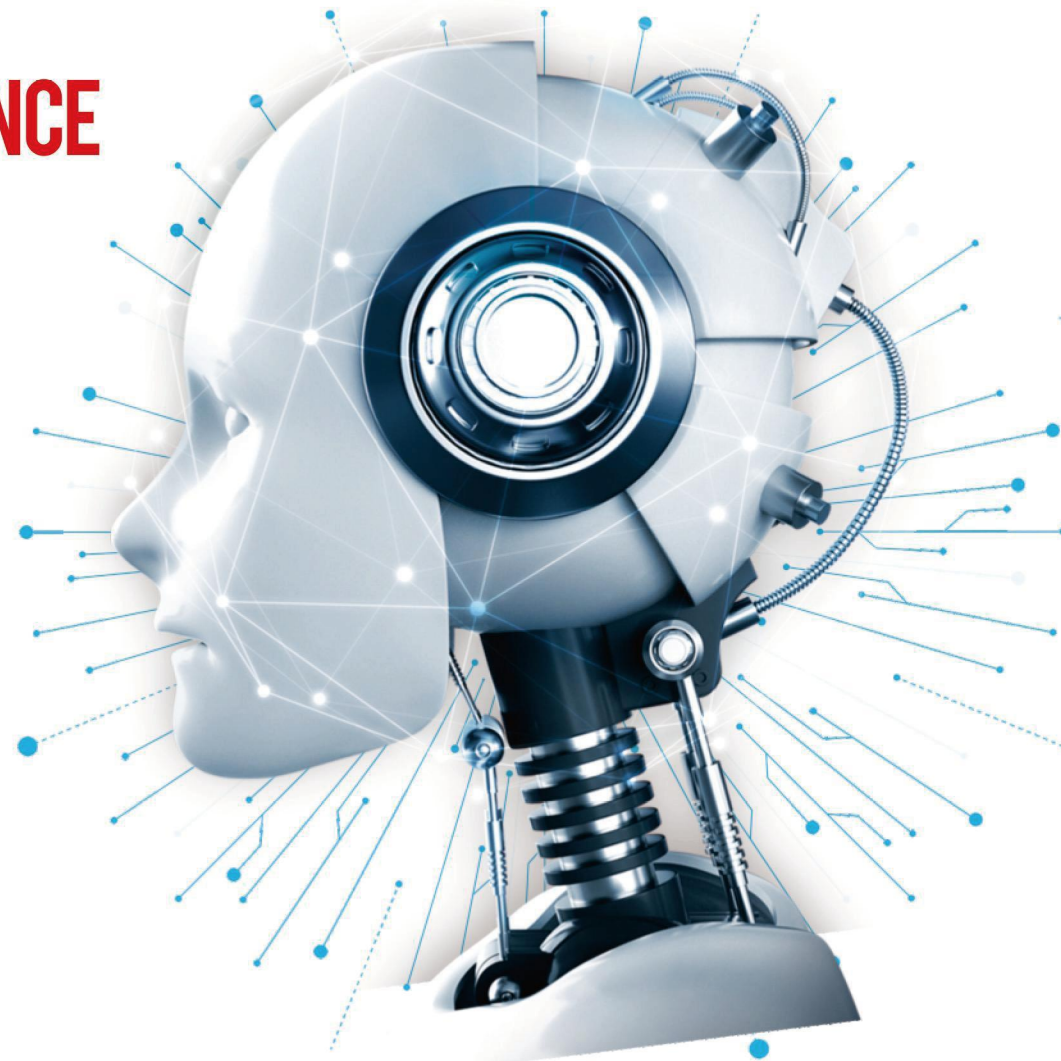
After the ore comes out of the silo it is fed into the separation channel by the electromagnetic vibratory feeder and flows into the detection area one piece after another. The detector performs qualitative and quantitative analysis on each pieces of ore, the computer analyzes the data by the set threshold and identifies which is valuable ore or gangue. At this point the valuable ore and tailings will be kicked to different tanks by the actuator.



INTELLIGENCE SYSTEM

Sparing no effort LONGi, continuously, seeks new breakthroughs of the present technology, and keeps taking the technology to new levels.

AI intelligent algorithm can realize self-inspect the working of machine and monitor the operation in real time.



Deep Learning Algorithm

Deep learning is machine learning. It's motivation is to establish and simulate a neural network for human brain analysis and learning, and to mimic the mechanism of the human brain to interpret data, such as images, sounds, and text.

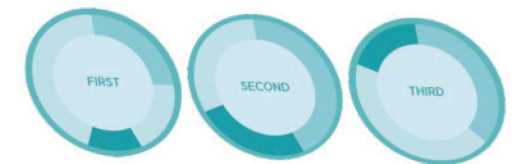
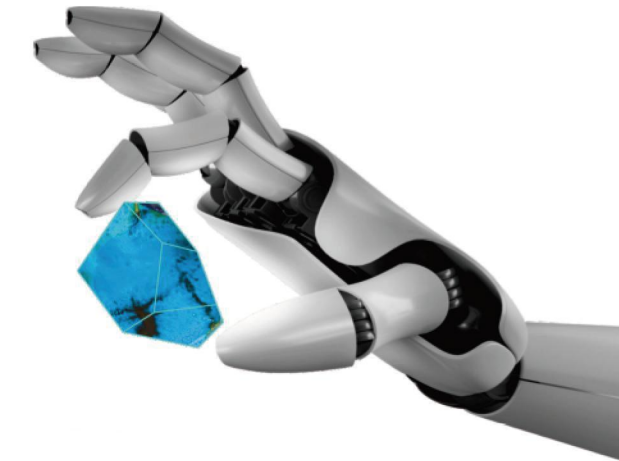
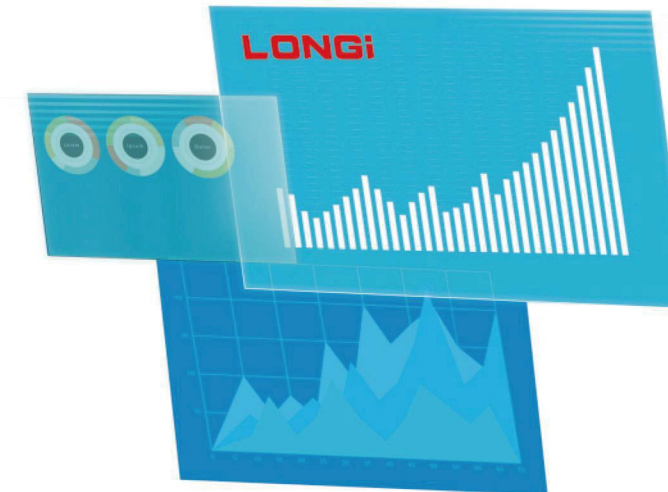
Deep learning is an artificial intelligence function that imitates the workings of the human brain in processing data and creating patterns for use in decision making.



Neural Network

The neural network is inspired by the structure of the brain. The neural network contains highly interconnected entities, called units or nodes. Neural networks are deep learning technologies. It generally focuses on solving complex processes. The neural network is a group of algorithms, these algorithms model the data using neurons for machine learning. It has the ability for self-learning and self-adaptation

Big Data Signal Processing - DAQ



The signal processing system is the central nervous system of the ore sorting machine, with the functions of data analysis and decision making.

The signal processing system processes the feature information acquired by the detection system at a high speed to form the detection image of the ore block, and carries out noise processing on the graphics or data. The analysis results obtained by combining the neural network and the intelligent deep learning algorithm are compared with the predetermined threshold values to complete the waste rock and ore identification.

Database For All Range Of Ore



PRODUCT QUALITY

LONGi holds over 341 national and international patents and multiple creative technological achievements;

Won the Chinese Patent Excellence Award in 2011

341

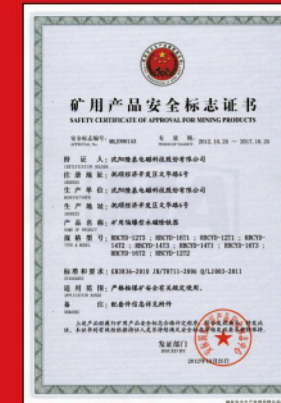
Won the national patent technology



CE Certification ISO9001 Quality Certification



Safety Certification and PCCC



Radiation Safety Permit

The safety protection of the LPNC and the LPPC separator was designed by a professional ray protection expert, and it is well protected by a sealed lead shell. The radiation within 1cm from the equipment is less than 2.0usv /h.

Eg. Calculating that the staff works 8 hours a day and 300 days a year near the equipment, it can be concluded that the annual radiation exposure = $2.0 \times 8 \times 300 \div 1000 = 4.8\text{mSv}$, which is much lower than the international criterion i.e the annual radiation exposure should not be higher than 20mSv.



TECHNICAL SPECIFICATIONS

LPNC(XRT) Intelligent Sorter



Model	LPNC12-80	LPNC22-80
Granularity Range	-80+15mm	-80+15mm
Capacity	>50t/h	>100t/h
Power	<16.0kw/h	<25.0kw/h
Weight	11t	20.0t
Outline Dimensions (L×D×H)	9200×1780×2500	9200×2780×2500

Note: Due to the different minerals different density and granularity properties, the processing capacity in the above table is for reference only.

Control System	
Model	LPNC-ZK-12-80
Power Required	220V 50Hz
Outline Dimensions (L×D×H)	1100X 800 X 1800
Weight	350Kg



LPPC(XRF) Intelligent Sorter



Model	LPPC1 - 150 (Special for Laboratory)	LPPC4 - 150
Particle Size	-150+30mm	-150+60mm
Capacity	1-1.5t/h	25-35t/h
The Number Of Separation Channels	1	4
Power Required	220V 50Hz	380V 50Hz
Power	1Kw	≤5Kw
Outline Dimensions (L×D×H)	2612 X 798 X 1970	5425 X 1786 X 2870
Weight	420Kg	5130Kg

Control System	
Model	LPPC-ZK-4-150
Power Required	220V 50Hz
Outline Dimensions (L×D×H)	1200X850X1300mm
Weight	110Kg



Test Data

The Site And Type Of The Ore	Size (mm)	Product	Yield%	Grade %	Recovery %
Inner Mongolia (Copper Ore)	-150+30	Concentrate	42.83	2.93	98.12
		Tailings	57.17	0.042	1.88
		Feeding	100.00	1.279	100.00
Inner Mongolia (Copper Ore)	-150+50	Concentrate	66.64	2.83	98.83
		Tailings	33.36	0.067	1.17
		Feeding	100.00	1.908	100.00
Xin Jiang (Copper Ore)	-60+20	Concentrate	58.22	0.484	93.49
		Tailings	41.78	0.047	6.51
		Feeding	100.00	0.301	100.00
Yunnan (Scheelite Ore)	-150+30	Concentrate	71.28	0.438	91.83
		Tailings	28.72	0.098	8.17
		Feeding	100.00	0.340	100.00
Jiangxi (Scheelite Ore)	-60+12	Concentrate	36.05	0.195	70.05
		Tailings	63.95	0.047	29.95
		Feeding	100.00	0.100	100.00
Inner Mongolia (Gold Ore)	-200+30	Concentrate	75.23	4.70g/t	98.22
		Tailings	24.77	0.25g/t	1.78
		Feeding	100.00	3.60g/t	100.00
Guangxi (Antimony Ore)	-90+60	Concentrate	15.97	22.47	99.12
		Tailings	84.03	0.038	0.88
		Feeding	100.00	3.62	100.00
Guangxi (Molybdenum Ore)	-100+30	Concentrate	50.83	0.095	86.68
		Tailings	49.17	0.015	13.32
		Feeding	100.00	0.056	100.00

The Site And Type Of The Ore	Size (mm)	Product	Yield%	Grade %		Recovery %	
Yunnan (Copper-zinc Ore)	-90+60			Cu	Zn	Cu	Zn
		Concentrate	20.19	1.389	2.411	86.29	88.51
		Tailings	79.81	0.056	0.079	13.71	11.49
		Feeding	100.00	0.325	0.550	100.00	100.00
Guangxi (Copper-zinc Ore)	-60+20			Cu	Zn	Cu	Zn
		Concentrate	65.43	0.027	2. 07	98.08	95.56
		Tailings	34.57	0.001	0.182	1.92	4.44
		Feeding	100.00	0.018	1.417	100.00	100.00
Hunan (Tungsten-molybdenum Ore)	-300+30			Wo3	Mo	Wo3	Mo
		Concentrate	36.92	0.25	0.056	82.98	78.45
		Tailings	63.08	0.03	0.009	17.02	21.55
		Feeding	100.00	0.111	0.026	100.00	100.00
Jiangxi (Tungsten-copper Ore)	-120+45			Wo3	Cu	Wo3	Cu
		Concentrate	29.49	0.800	0.485	91.44	86.16
		Tailings	70.51	0.031	0.033	8.56	13.84
		Feeding	100.00	0.258	0.166	100.00	100.00
Inner mongolia (Lead-zinc Ore)	-150+30			Zn	Pb	Zn	Pb
		Concentrate	51.54	5.910	1.346	98.48	97.02
		Tailings	48.46	0.097	0.044	1.52	2.98
		Feeding	100.00	3.093	0.715	100.00	100.00
Yunnan (Lead-zinc Ore)	-150+30			Zn	Pb	Zn	Pb
		Concentrate	70.05	7.520	5.436	99.90	99.71
		Tailings	29.95	0.017	0.038	0.10	0.29
		Feeding	100.00	5.273	3.819	100.00	100.00

From above the test data, we can see that most of the waste can be thrown away by the intelligent sorter in order to improve the ore grade that will be feed into the processing line so as to decrease the cost of subsequent crushing and grinding and processing. The pre-processing by the intelligent sorter saves energy and the operation costs which is very helpful to improve the economic benefit of the processing plant. The equipment runs stably in wide variety of applications. The intelligent sorter is going to be an indispensable piece of equipment for mineral processing.

Pilot Test



Achievements



GLOBAL PARTNERSHIP

LONGi products are well known all over the world.
LONGi achieves great recognition and praise from customers.
LONGi always appreciates the customers' feedback
LONGi always strives to meet the customer's demands.

3000

We have provided solutions for more than 3,000 customer worldwide.

30

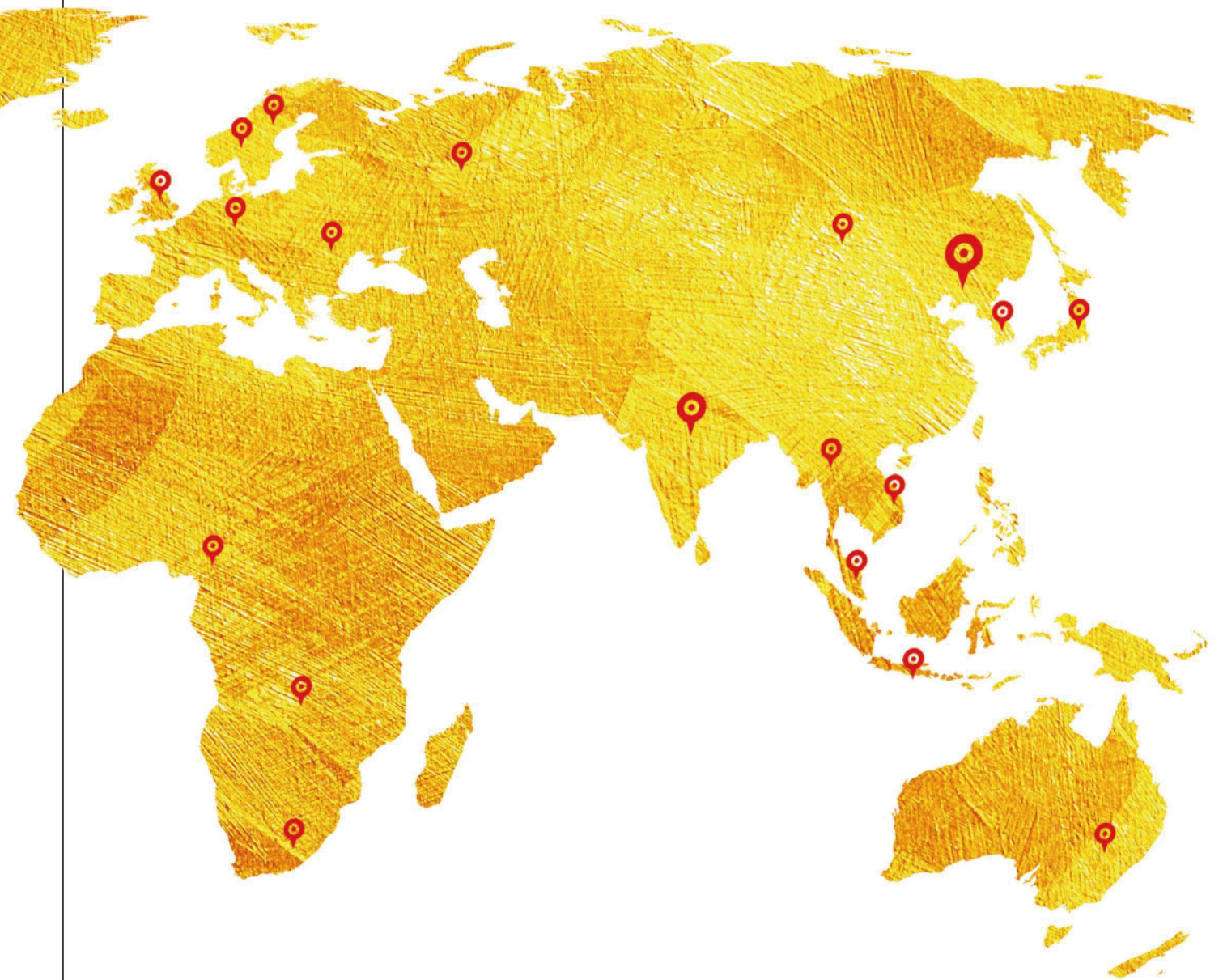
Product exports to more than 30 countries

The Partners All Over The World.

LONGi, as the expert of the processing technology, always practices the long term service promise. "Focus on customer's request, stay close to the market, offer supply system solutions and keep improving the value of our products are our missions.

Building a closed matrix service system covers the equipment performance, on-site investigation, material analysis, metallurgical experimentation, standard customization, solidwork dynamic simulation analysis and virtual installation, installation and debugging training service, big data maintenance system tracking, among others.

Integrating various servicing behaviors from customer's procurement, equipment operation to the long term technical support, LONGi has built a value chain based on customer demand and oriented to accommodate customer requests from all over the world.





QUALITY AND SERVICE

LONGi looks forward to a commitment to customers with quality and service as the life of the company and the technology and production as the soul of the company.

LONGi performs the international standard ISO9001 QS systems, provides advanced inspection methods for improving our quality continuously.

LONGi also provides remote service to our customer, solves the after-sales issues through the networking and data transmission.

48
Quick Response In
48 Hours



COMPREHENSIVE SOLUTIONS

LONGi provides comprehensive solutions to save customers' time and money.

LONGi always responds to customers when they need us at the first time, and propose the best solutions within 48 hours.

